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Cancers Among Adolescents and Young Adults in Maine 2001–2017: A Descriptive Analysis and Model for Future Studies

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Purpose: Maine has had a high age-adjusted incidence rate of cancer and a high incidence rate of childhood cancer. However, it is unknown if Maine adolescents and young adults (AYAs) have a higher-than-expected cancer incidence rate.

Methods: Based on the International Classification of Diseases of Oncology, Third Edition (ICD-0-3) classification system, we obtained the incidence rates of AYA cancers by state, sex, race, and ethnicity from 2001 to 2017 using CDC WONDER. Overall, all cancer rates and age-adjusted incidence rates for specific cancers were compared and ranked by state.

Results: Maine had a similar overall age-adjusted cancer incidence rate among non-Hispanic white AYAs compared with the national rate from 2001 to 2017. Among non-Hispanic white AYA cancers of the respiratory system, lung and bronchial cancers, uterine cancer, female breast cancer, thyroid cancer, lymphomas, and Hodgkin's lymphoma were significantly greater in Maine than the state with the lowest incidence rate for that cancer. Specifically, Maine was ranked as one of the top 10 states for the highest incidence rate of lung cancer, uterine cancer, Hodgkin's lymphoma, thyroid cancer (males only), and female breast cancer.

Conclusion: Although the cancer incidence rate among AYAs in Maine was not significantly higher than the national average, Maine AYAs had a significantly higher incidence of lung cancer, uterine cancer, Hodgkin's lymphoma, thyroid cancer, and breast cancer. This information may direct future studies to identify regional cancer risk factors in Maine, which may account for this elevated rate. The described methodology is an innovative approach to identifying states with high cancer rates and directing future studies.

Keywords: AYA, cancer, incidence rate, Maine, adolescents, young adults

Introduction

OVER THE LAST two decades, Maine has consistently had a higher age-adjusted cancer incidence rate than the national average. This high rate has been primarily a reflection of the high rates of lung cancer, bladder cancer, cancer of the oral cavity, non-Hodgkin's lymphoma (males), and uterine cancer in Maine.^{1,2} In addition, the incidence of childhood cancer is also elevated in Maine. Based on data from 2003 to 2014, Maine had the third-highest incidence rate of childhood cancer.³ Although it is unclear why Maine has an elevated rate of cancer, Maine has several unique cancer risk factors. The rocks and soils of Maine create more radon than most other

states, Maine has a higher-than-average number of smokers, and many wells in Maine have high arsenic levels.^{4–6} All of these factors are associated with an increased risk of cancer. The incidence rate of cancers among adolescents and young adults (AYAs) age 15–39 years has not been described in Maine. It is unknown if AYAs in Maine have a higher-than-expected incidence of cancer, as do their older and younger counterparts. To provide a complete picture of the cancer incidence in Maine, we performed a descriptive analysis of cancers that occurred among AYAs in Maine from 2001 to 2017 using CDC WONDER. To highlight cancers that have a high incidence rate, we compared incidence rates of cancer among AYAs in Maine to other states in the nation.

Results

Of the 5391 total cancers identified among AYAs in Maine, we identified 5036 cancers (93.1%) among non-Hispanic whites from 2001 to 2017. Maine AYAs had a similar overall cancer incidence rate (81.3/100,000) to the national average (80.0/100,000) (Table 1). This rate increased over time. From 2001 to 2017, the cancer incidence rate among AYAs in Maine increased by 23.2% (74.1 to 91.3/100,000 persons) and was similar to the 10.5% increase among AYAs in the nation (73.6 to 81.4/100,000 persons) (Fig. 1). AYA cancer rates in Maine also increased by age and varied by sex (Fig. 2). The overall cancer incidence rate among female AYAs was 99.9 per 100,000 and was higher than the overall rate among males at 62.1 per 100,000. This difference was primarily due to higher rates of thyroid and breast cancer among females. Males had a higher incidence for all cancer sites, except melanoma, thyroid, and breast cancers (Tables 2 and 3).

Compared with the national average, Maine male and female AYAs had a higher rate of cancers of the respiratory system (2.3 vs. 1.8/100,000), lung and bronchial cancers (1.9 vs. 1.4/100,000), thyroid cancer (12.7 vs. 11.5/100,000), and Hodgkin's lymphoma (5.1 vs. 4.4/100,000). Maine AYAs had a lower rate of melanoma (9.4 vs. 10.5/100,000) than the national average. When compared with the states with the lowest incidence rate for each cancer site, Maine AYAs had a higher incidence rate of cancers at all sites (81.3 vs. 69.5/100,000), cancers of the digestive system (6.7 vs. 4.9/100,000), cancers of the respiratory system (2.3 vs. 1.0/100,000), cancers of the lung and bronchus (1.9 vs. 0.8/100,000), melanoma (9.4 vs. 6.6/100,000), thyroid cancer (12.7 vs. 7.6/100,000), and lymphoma (9.0 vs. 6.7/100,000).

When ordinal ranked, Maine was ranked the following: all sites (13th), cancers of the digestive tract (13th), cancers of the respiratory system (2nd), cancers of the lung and bronchus (2nd), melanoma (34th), thyroid cancer (10th), and lymphoma (7th) (Table 1). Although Maine was ranked second for pancreatic cancer, fifth for cancers of the bones and joints, and third for cancers of the brain, Maine's rates for these cancer sites were considered similar to the state with the lowest incidence rate for each of these cancers as their confidence intervals overlapped.

Male AYAs in Maine had a similar overall incidence rate of cancer (62.1/100,000) compared with the national average (62.2/100,000). Cancer rates by cancer site were also similar to the national average. However, when compared with the states with the lowest incidence rate of cancer, AYA males in Maine had higher incidence rates of cancer at all sites (62.1 vs. 53.7/100,000), cancers of the respiratory system (2.2 vs. 1.1/100,000), lung and bronchial cancers (1.7 vs. 0.7/100,000), Hodgkin's lymphoma (5.4 vs. 3.2/100,000), testicular cancer (13.4 vs. 9.9/100,000), and thyroid cancer (4.8 vs. 2.6/100,000). When ordinal ranked, Maine was ranked the following: all sites (17th) respiratory cancers (6th), cancers of the lung and bronchus (2nd), Hodgkin's lymphoma (6th), testicular cancer (21st), and thyroid cancer (7th) (Table 2).

Female AYAs in Maine had a similar overall incidence rate of cancer compared with the national average (99.9 vs. 98.1/100,000). However, when compared by cancer site, female AYAs had a higher incidence of cancers of the respiratory system (2.4 vs. 1.7/100,000), lung and bronchus

CDC WONDER (Wide-Ranging Online Data for Epidemiologic Research) is a publicly available online information tool, which provides access to the United States Cancer Statistics (USCS). The USCS comprises data obtained from the National Program of Cancer Registries (NPCR) and the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) program and represents 97% of the U.S. population.^{5,7} In comparison, SEER data, upon which prior AYA and Maine cancer incidence reports were based, covers only 35% of the United States and limits comparisons between states.^{2,8–10}

Methods

We obtained the incidence rates of AYA cancers by state, sex, race, and ethnicity from 2001 to 2017 from the USCS using CDC WONDER.¹¹ Only cases of invasive cancer in individuals between 15 and 39 years of age were included for analysis. Invasive cancers were described by site according to the International Classification of Diseases of Oncology, Third Edition (ICD-0-3).¹² Because ICD-0-3 was implemented in 2000, only data from 2001 and later was used in our analysis.

Cancers were reported by group and subgroup. The digestive system site group included cancers of the esophagus, stomach, small intestine, colon, rectum, anus, liver, gallbladder, pancreas, retroperitoneum, and peritoneum. Cancers of the respiratory system site group included cancers of the nose, ear, larynx, lung, bronchus, trachea, and mediastinum. Cancer of the urinary system site group included cancer of the bladder and kidney. Lymphomas included Hodgkin's lymphoma and non-Hodgkin's lymphoma. Leukemia included both acute and chronic leukemias. Female genital system site group cancers included cervical, uterine, ovarian, vaginal, and vulvar cancers.

Age group-specific incidence rates, age-adjusted incidence rates, and 95% confidence intervals by sex, race, ethnicity, and state were obtained from the CDC WONDER platform. Age-adjusted incidence rates with 95% confidence intervals were obtained by cancer groups and subgroups from the CDC WONDER platform. Age-adjusted incidence rates were considered significant if the 95% confidence interval for the two rates overlapped. Non-overlapping confidence intervals were considered significantly different and described as "higher" or "lower." To describe the incidence rate of cancer among AYAs in Maine relative to rates in other states, we rank-ordered the states by their age-adjusted incidence rates and assigned "1" to the state with the highest rate. To identify cancers that may be due to regional risk factors in Maine, the incidence rate of cancer among AYAs in Maine was compared with the state with the lowest incidence rate for that cancer.

Due to the low numbers of cancers among individuals identifying as non-white or Hispanic in Maine and because cancer incidence rates among AYAs vary by race and ethnicity, we limited our analysis to individuals identifying as non-Hispanic white.^{3,13,14}

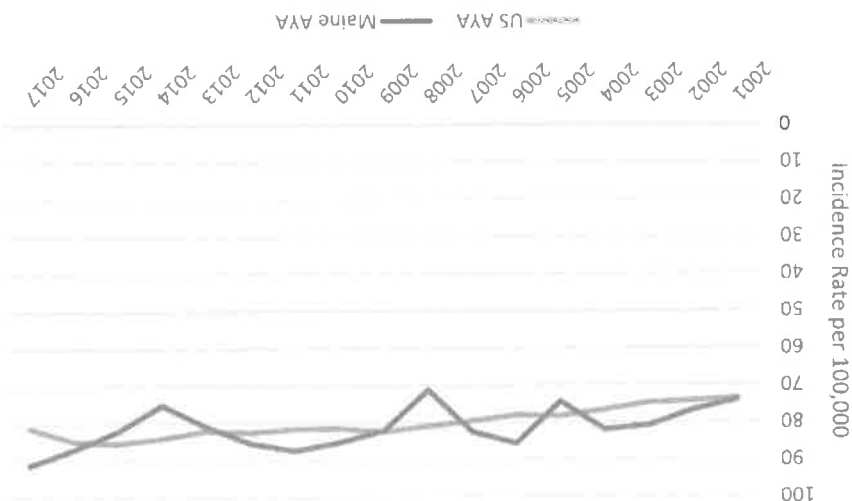
TABLE 1. AGE-ADJUSTED CANCER INCIDENCE RATE AND RANKING AMONG ADOLESCENTS AND YOUNG ADULTS IN MAINE, UNITED STATES, AND SELECTED STATES, 2001–2017

Cancer site	National rate ^a	Maine rate ^a	Maine ranking	Lower incidence rate by state ^a	Highest incidence rate by state ^a
All cancer sites combined	80.0 (79.9–80.2)	81.3 (79.1–83.6)	13	69.5 (66.4–72.8) Alaska	98.4 (93.5–103.4) Hawaii
Oral cavity and pharynx	1.5 (1.5–1.5)	1.5 (1.3–1.9)	20	1.2 (0.8–1.7) Wyoming	2.6 (1.9–3.5) Hawaii
Digestive system	6.4 (6.3–6.4)	6.7 (6.1–7.4)	13	4.9 (4.1–5.9) Wyoming	7.6 (7.0–8.2) West Virginia
Colon and rectum	4.2 (4.1–4.2)	4.3 (3.8–4.9)	16	3.2 (2.6–4.0) Wyoming	5.2 (4.7–5.7) West Virginia
Stomach	0.5 (0.4–0.5)	0.5 (0.3–0.7)	12	0.3 (0.3–0.4) Virginia	0.6 (0.5–0.7) New Jersey
Pancreas	0.5 (0.5–0.5)	0.6 (0.4–0.8)	2	0.3 (0.2–0.5) Nevada	0.6 (0.5–0.7) Louisiana
Respiratory system	1.8 (1.8–1.8)	2.3 (2.0–2.7)	2	1.0 (0.7–1.5) North Dakota	2.7 (2.4–3.1) West Virginia
Lung and bronchus	1.4 (1.3–1.4)	1.9 (1.5–2.2)	2	0.8 (0.6–0.9) Utah	2.1 (1.8–2.5) West Virginia
Bones and joints	0.9 (0.9–1.0)	0.9 (0.7–1.1)	22	0.6 (0.4–0.9) South Dakota	1.2 (1.1–1.3) Florida
Soft tissue, including heart	1.5 (1.5–1.6)	1.5 (1.2–1.8)	24	1.3 (1.2–1.5) Indiana	1.8 (1.7–2.0) New Jersey
Melanoma of the skin	10.5 (10.5–10.6)	9.4 (8.6–10.2)	34	6.6 (5.7–7.7) Alaska	22.7 (20.4–25.2) Hawaii
Urinary system	3.0 (3.0–3.1)	2.8 (2.3–3.2)	27	2.0 (1.6–2.5) New Mexico	4.3 (3.7–5.0) Rhode Island
Urinary bladder	0.9 (0.8–0.9)	0.8 (0.6–1.1)	24	0.5 (0.3–0.8) New Mexico	1.4 (1.0–1.8) Rhode Island
Kidney and renal pelvis	2.2 (2.2–2.2)	2.0 (1.6–2.3)	24	1.4 (1.0–1.8) North Dakota	3.1 (2.7–3.4) Arkansas
Brain and other nervous system	3.7 (3.7–3.7)	3.9 (3.4–4.4)	9	3.0 (2.4–3.6) South Dakota	4.1 (3.9–4.4) New Jersey
Thyroid	11.5 (11.4–11.6)	12.7 (11.8–13.6)	10	7.6 (7.1–8.1) Arkansas	17.3 (16.8–17.8) New Jersey
Lymphomas	8.4 (8.4–8.5)	9.0 (8.2–9.7)	7	6.7 (5.8–7.8) Wyoming	11.5 (11.1–11.9) New Jersey
Hodgkin's lymphoma	4.4 (4.4–4.5)	5.1 (4.6–5.7)	5	3.4 (3.0–3.8) Mississippi	6.0 (5.7–6.3) New Jersey
Non-Hodgkin's lymphoma	4.0 (4.0–4.0)	3.9 (3.4–4.4)	17	3.1 (2.5–3.9) Wyoming	5.5 (5.2–5.8) New Jersey
Leukemia	3.1 (3.1–3.2)	2.8 (2.4–3.2)	40	2.5 (1.9–3.2) Wyoming	4.0 (3.4–4.8) South Dakota
Acute lymphocytic leukemia	0.8 (0.8–0.8)	0.7 (0.5–0.9)	33	0.6 (0.5–0.7) Virginia	1.0 (0.9–1.1) Florida
Acute myeloid leukemia	1.2 (1.2–1.2)	1.2 (0.9–1.5)	18	0.7 (0.4–1.2) Wyoming	1.8 (1.5–2.1) West Virginia

U.S. numbers do not include Massachusetts, Delaware, Kansas, Kentucky, or Pennsylvania; Rates for non-Hispanic whites.

^aRates per 100,000 AYA population with 95% confidence interval.
AYA, adolescents and young adults.

FIG. 1. Age-adjusted incidence rate of AYA cancer in Maine and the United States, 2001-2017. Incidence rates for non-Hispanic whites, AYAs, adolescents and young adults.



Discussion

This is the first report to describe cancer incidence rates among AYAs in Maine and identify cancers that are elevated among AYAs by comparing incidence rates to other states using CDC WONDER.

The County Health Rankings & Roadmaps program,¹⁵ This program compares health indicators between counties with the goal to increase awareness of health disparities between counties and highlight areas for possible improvement. Based on this methodology, we showed that the incidence rate of cancer among AYAs differed substantially by state and that Maine had elevated rates of lung cancer, uterine cancer, Hodgkin's lymphoma, breast cancer, testicular cancer, and melanoma among AYAs, suggesting the presence of cancer risk factors unique to Maine.

Maine was ranked as one of the top five states with the highest incidence rates for cancers of the lung, bronchus, and uterus and was one of the top 10 states with the highest

incidence intervals overlapped (Table 3). lowest incidence rate for each of these cancers as their counterparts were considered similar to the state with the lowest incidence rate for each of these cancers as their counterparts were ranked first and fourth, respectively, pancreatic cancer were ranked first and fourth, respectively, (9th), and Hodgkin's lymphoma (7th). Although stomach and system (22nd), corpus uteri (3rd), thyroid (14th), lymphomas (1st), melanoma (33rd), breast cancer (9th), female genital sites (13th), respiratory system (2nd), lung and bronchus When ordinal ranked, Maine was ranked the following: all 3.1/100,000). (8.1 vs. 5.7/100,000); and Hodgkin's lymphoma (4.8 vs. 100,000); thyroid cancer (12.6 vs. 10.4/100,000); lymphoma (100,000); cancer of the uterine corpus (4.0 vs. 2.0/100,000); cancer of the female genital system (13.8 vs. 10.4/100,000); (6.6 vs. 4.2/100,000); melanoma (11.9 vs. 8.3/100,000); cancers of the all sites (99.9 vs. 84.8/100,000); breast cancer (22.6 vs. 17.8/100,000). AYAs in Maine also had a higher incidence rate of cancer at with the lowest incidence rate for each cancer site, female (2.0 vs. 1.4/100,000), and corpus uteri (4.1 vs. 3.0/100,000)

FIG. 2. Crude cancer incidence rate among non-Hispanic white AYAs by sex and age, Maine, 2001-2017.

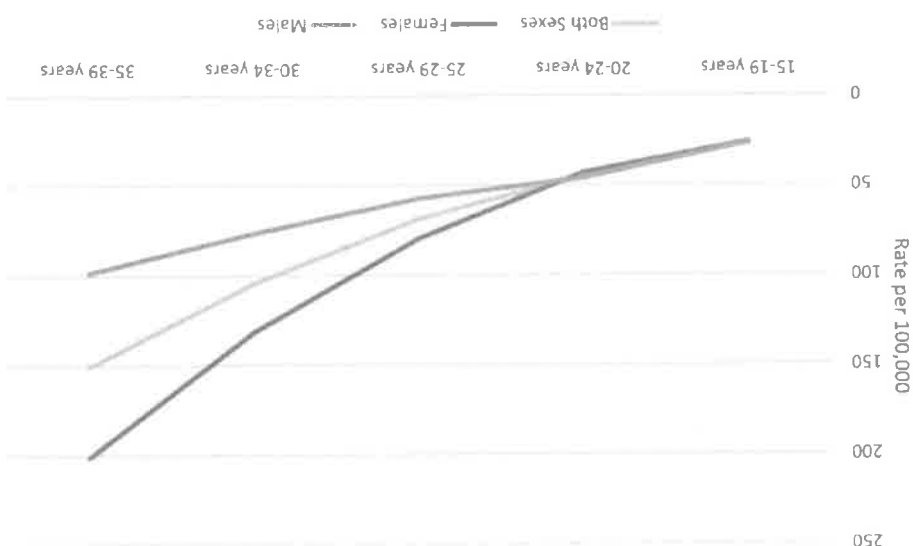


TABLE 2. AGE-ADJUSTED CANCER INCIDENCE RATE AND RANKING AMONG MALE ADOLESCENTS AND YOUNG ADULTS IN MAINE, UNITED STATES, AND SELECTED STATES, 2001–2017

Cancer site	National male AYA incidence rate of cancer ^a	Maine male AYA incidence rate of cancer ^a	Maine male AYA cancer incidence rate ranking	Lowest incidence rate of male AYA cancer by state ^a	Highest incidence rate of male AYA cancer by state ^a
All cancer sites combined	62.2 (62.0–62.4) ^a	62.1 (59.4–65.0)	17	53.7 (52.0–55.4) Oklahoma	81.3 (75.4–87.6) Hawaii
Oral cavity and pharynx	1.7 (1.7–1.7)	1.8 (1.3–2.3)	12	1.3 (1.1–1.6) Utah	2.9 (1.8–4.2) Hawaii
Digestive system Colon and rectum	6.6 (6.6–6.7) 4.3 (4.2–4.3)	6.9 (6.0–7.9) 4.4 (3.7–5.2)	13 17	5.1 (4.4–5.9) Nevada 3.2 (2.3–4.4) Wyoming	8.0 (7.1–9.0) Mississippi 5.4 (4.6–6.1) Mississippi
Stomach and pancreas	0.5 (0.5–0.5) 0.3 (0.2–0.5)	NA NA	NA NA	0.3 (0.2–0.4) Virginia 0.3 (0.2–0.5) Utah	0.9 (0.5–1.4) New Mexico 0.7 (0.5–1.0) Mississippi
Respiratory system Lung and bronchus	1.9 (1.8–1.9) 1.3 (1.2–1.3)	2.2 (1.7–2.8) 1.7 (1.3–2.2)	6 2	1.1 (0.7–1.6) Montana 0.7 (0.5–1.0) Utah	3.0 (2.5–3.6) West Virginia 2.3 (1.8–2.8) West Virginia
Bones and joints Soft tissue, including heart Melanoma of the skin	1.3 (1.1–1.5) 1.7 (1.6–1.7) 7.7 (7.6–7.7)	1.3 (0.9–1.7) 1.4 (1.0–1.9) 6.8 (5.9–7.8)	5 42 33	0.7 (0.5–1.1) Mississippi 2.3 (1.6–3.0) New Mexico 5.1 (4.0–6.4) Alaska	1.6 (1.3–1.9) Utah 1.3 (1.1–1.5) Indiana 19.7 (16.8–23.0) Hawaii
Testes	13.0 (12.9–13.1)	13.4 (12.1–14.7)	21	9.9 (9.2–10.7) Oklahoma	16.9 (14.3–19.9) Hawaii
Urinary system Urinary bladder Kidney and renal pelvis	3.7 (3.7–3.8) 1.2 (1.2–1.2) 2.5 (2.5–2.6)	3.1 (2.5–3.8) 1.1 (0.8–1.6) 2.0 (1.5–2.6)	39 24 38	2.5 (2.2–2.8) Colorado 0.8 (0.5–1.3) New Mexico 1.7 (1.5–2.0) Colorado	5.4 (4.7–6.2) West Virginia 2.0 (1.2–3.3) Hawaii 3.9 (3.3–4.5) West Virginia
Brain and other Nervous system Thyroid	4.1 (4.1–4.2) 4.2 (4.1–4.2)	4.5 (3.8–5.3) 4.8 (4.1–5.6)	3 7	3.5 (2.7–4.5) North Dakota 2.6 (2.2–3.0) Arkansas	4.8 (4.1–5.5) Idaho 6.9 (6.4–7.3) New Jersey
Lymphomas Hodgkin's Lymphoma Non-Hodgkin's lymphoma	9.3 (9.3–9.4) 4.5 (4.5–4.6) 4.8 (4.7–4.9)	9.9 (8.8–11) 5.4 (4.6–6.2) 4.5 (3.8–5.3)	12 6 22	7.5 (6.1–9.1) Wyoming 3.2 (2.7–3.9) Nevada 3.5 (2.7–4.5) South Dakota	12.5 (11.9–13.2) New Jersey 5.9 (5.5–6.4) New Jersey 6.6 (6.2–7.1) New Jersey
Leukemia Acute Lymphocytic leukemia Acute myeloid leukemia	3.6 (3.5–3.6) 0.9 (0.6–1.3) 1.2 (1.1–1.2)	3.1 (2.6–3.8) 0.9 (0.6–1.3) 1.2 (0.9–1.7)	42 32 18	2.8 (2.5–3.1) Virginia 0.7 (0.6–0.9) Virginia 0.8 (0.5–1.1) Nebraska	5.2 (4.2–6.3) South Dakota 1.4 (0.9–2.1) Vermont 1.7 (1.2–2.5) South Dakota

U.S. numbers do not include Massachusetts, Delaware, Kansas, Kentucky, or Pennsylvania. Rates for non-Hispanic whites.

^aRates per 100,000 AYA population with 95% confidence interval.

NA. Data not available through CDC WONDER.

TABLE 3. AGE-ADJUSTED CANCER INCIDENCE RATE AND RANKING AMONG FEMALE ADOLESCENTS AND YOUNG ADULTS IN MAINE, UNITED STATES, AND SELECTED STATES, 2001–2017

Cancer site	National female AYA ^a incidence rate of cancer ^a	Maine female AYA incidence rate of cancer ^a	Maine female AYA cancer incidence rate ranking	Lowest incidence rate of female AYA cancer by state ^a	Highest incidence rate of female AYA cancer by state ^a
All cancer sites combined	98.1 (97.8–98.3)	99.9 (96.5–103.5)	13	84.8 (79.8–90.1) Alaska	120.5 (112.4–129.0) Hawaii
Oral cavity and pharynx	1.3 (1.3–1.3)	1.4 (1.0–1.8)	13	1.0 (0.9–1.2) Virginia	2.3 (1.3–3.7) Hawaii
Digestive system Colon and rectum	6.1 (6.0–6.2) 4.1 (4.0–4.2)	6.6 (5.7–7.6) 4.2 (3.5–5.0)	10 15	4.7 (3.5–6.1) Wyoming 3.2 (2.3–4.5) Wyoming	7.6 (6.8–8.3) West Virginia 5.7 (4.0–7.8) Hawaii
Stomach and Pancreas	0.4 (0.4–0.4) 0.5 (0.5–0.5)	0.6 (0.3–0.9) 0.6 (0.4–1.0)	1 4	0.3 (0.2–0.4) Washington 0.3 (0.2–0.5) Arkansas	0.6 (0.3–0.9) Maine 0.7 (0.5–0.9) Maryland
Respiratory system	1.7 (1.7–1.8)	2.4 (1.9–3.0)	2	1.1 (0.9–1.4) Utah	2.5 (2.0–3.0) West Virginia
Lung and bronchus	1.4 (1.4–1.5)	2.0 (1.6–2.6)	1	0.8 (0.6–1.1) Utah	2.0 (1.6–2.6) Maine
Bones and joints	0.8 (0.8–0.8)	NA	NA	0.6 (0.4–0.9) Nevada	1.1 (0.9–1.6) New Hampshire
Soft tissue, including heart	1.4 (1.4–1.4)	1.5 (1.1–2.0)	10	1.0 (0.6–1.7) North Dakota	1.8 (1.1–2.6) Vermont
Melanoma of the skin	13.4 (13.3–13.5)	11.9 (10.7–13.2)	33	8.3 (6.8–10.1) Alaska	27.0 (23.3–31.1) Hawaii
Female breast	22.1 (22.0–22.3)	22.6 (21.0–24.4)	9	17.8 (15.4–20.4) Wyoming	27.5 (26.6–28.4) New Jersey
Female genital system	13.7 (13.6–13.8)	13.8 (12.6–15.2)	22	10.4 (9.6–11.3) Connecticut	21 (19.7–22.4) West Virginia
Cervix uteri	6.9 (6.8–6.9)	6.6 (5.7–7.6)	25	4.2 (3.7–4.6) Utah	9.2 (8.3–10.1) West Virginia
Corpus uteri	3.1 (3.0–3.1)	4.0 (3.3–4.7)	3	2.0 (1.8–2.3) Colorado	6.4 (5.7–7.2) West Virginia
Ovary	2.9 (2.9–3.0)	2.5 (2.0–3.2)	34	1.9 (1.4–2.6) Montana	4.0 (2.6–5.8) Hawaii
Urinary system	2.4 (2.3–2.4)	2.4 (1.9–3.0)	16	1.4 (0.9–2.0) New Mexico	3.5 (2.7–4.4) Rhode Island
Urinary bladder	0.5 (0.5–0.5)	NA	NA	0.3 (0.3–0.5) Virginia	0.9 (0.6–1.4) Rhode Island
Kidney and renal pelvis	1.8 (1.8–1.9)	1.9 (1.4–4)	17	1.1 (0.9–1.4) Colorado	2.7 (2.3–3.2) Arkansas
Brain and other nervous system	3.2 (3.2–3.3)	3.1 (2.5–3.8)	24	2.2 (1.6–3.1) South Dakota	3.9 (2.5–5.7) Hawaii
Thyroid	18.9 (18.8–19.0)	20.4 (18.9–22.1)	14	12.6 (11.7–13.6) Arkansas	27.8 (26.9–28.8) New Jersey
Lymphomas	7.5 (7.4–7.6)	8.1 (7.1–9.1)	9	5.7 (5.1–6.4) Arkansas	10.4 (9.9–11.0) New Jersey
Hodgkin's lymphoma	4.3 (4.3–4.4)	4.8 (4.1–5.6)	7	3.1 (2.5–3.7) Mississippi	6.1 (5.7–6.6) New Jersey
Non-Hodgkin's lymphoma	3.2 (3.1–3.2)	3.3 (2.7–4.0)	12	2.2 (1.4–2.7) Wyoming	4.3 (4.0–4.7) New Jersey
Leukemia	2.7 (2.6–2.7)	2.4 (1.9–3.0)	35	1.5 (0.9–2.4) Wyoming	4.0 (2.6–5.9) Hawaii
Acute lymphocytic leukemia	0.5 (0.5–0.6)	NA	NA	0.4 (0.3–0.6) Tennessee	0.8 (0.6–1.0) Connecticut
Acute myeloid leukemia	1.2 (1.2–1.2)	1.2 (0.8–1.6)	18	0.9 (0.6–1.3) Nebraska	1.9 (1.5–2.3) West Virginia

U.S. numbers do not include Massachusetts, Delaware, Kansas, Kentucky, or Pennsylvania; Rates for non-Hispanic whites.

^aRates per 100,000 AYA population with 95% confidence interval.

NA. Data not available through CDC WONDER.

incidence rates of Hodgkin's lymphoma, breast cancer, and thyroid cancer. The high rate of cancers of the lung and bronchus among AYAs in Maine is consistent with the high rate of lung cancer among older adults in Maine, suggesting that both age groups may be affected by similar risk factors.^{1,16} Although Maine has unique risk factors for lung cancer, such as a higher-than-average smoking rate and areas with high levels of radon, it is unclear if these risk factors account for the high rate of lung cancer.⁴ It has been speculated that there are additional risk factors unique to Maine that may account for this elevated rate.^{17,18}

Maine's high rate of uterine cancer among AYAs is consistent with rates among older women in Maine, which suggests the presence of risk factors unique to Maine. Risk factors for uterine cancer include obesity, genetics, diabetes, a diet high in animal fat, extended exposure to estrogens, and treatment of prior cancers.¹⁹ It is unclear if Maine has a higher incidence of these risk factors.

The incidence of cancers among AYAs is increasing, and it is unclear as to what risk factors account for this increase.⁹ There is limited literature regarding risk factors for AYA cancers.²⁰ Most cancer risk factors are based on evidence reflecting the older population. Considering that cancers among AYAs are unique and incompletely understood, AYA cancers may also be the result of risk factors that have not been identified.^{21,22} As a first step to understanding the epidemiology of AYA cancers, temporal-spatial clustering of AYA cancers should be performed. Cluster analysis and identification of cancer risk factors may ultimately be easier in the AYA population. Cancer rates among AYAs are less likely to be influenced by cancer screening rates because cancer screenings typically start after age 45 (excluding cervical cancer screening) and the AYA population has fewer lifetime carcinogenic exposures.²³

There are several limitations of this study. This report describes cancer among non-Hispanic white AYAs and does not represent cancers that occur among AYAs of other races and ethnicities.

Access to health care may be a confounding factor and influence cancer rates among AYA individuals. Specifically, thyroid and skin cancers can be indolent, and diagnosis may be made as an incidental finding on clinical exam or in the workup for an unrelated health condition. In addition, the incidence of cervical cancer is associated with the rate of cervical cancer screening. Therefore, the incidence rates of thyroid cancer, melanoma, and cervical cancer may be related to access to health care.

Registry data quality may have influenced data for this study. The completeness and quality of state-specific data may vary and influence rates. Although the cancer incidence rates were available for all states, data by ethnicity were not available for Pennsylvania, Massachusetts, Kansas, Kentucky, and Delaware. Although the absence of complete data from these states may slightly change the national average for cancer incidence by ethnicity, it does not change our conclusion that Maine has a significantly higher incidence rate of specific cancers among AYAs.

We were unable to calculate *p*-values or higher statistical analysis because CDC WONDER does not provide the original data. Defining differences in cancer rates by overlapping or nonoverlapping confidence intervals may underestimate the number of cancers with significantly different

Conclusion

This report describes and compares Maine AYA's cancer incident rate to other states to identify cancers that occur at a higher rate and may benefit from an epidemiologic study. Compared with the national average, Maine AYAs have a higher incidence rate of lung and bronchial cancers, Hodgkin's lymphoma, thyroid cancer, and uterine cancer. AYAs in Maine also have an elevated incidence of breast cancer, testicular cancer, and melanoma compared with states with a lower incidence of cancer. The elevated incidence rates of these cancers suggest the presence of regional risk factors in Maine. If risk factors are identified, these cancers may be amenable to modification through primary preventive measures.

Authors' Contributions

The author confirms sole responsibility for the following: study conception, design, data collection, analysis and interpretation of results, and preparation of article.

Acknowledgment

The author acknowledges the helpful input of R. Gibson Parrish, MD, in preparing this article.

Author Disclosure Statement

No competing financial interests exist.

Funding Information

No funding was received for this article.

References

- Yob D, Haggan K, Bancroft C, et al. The Maine 2020 annual report of Cancer. Augusta, ME: Maine CDC Cancer Registry; 2021.
- Maine Center for Disease Control & Prevention. Main Cancer Registry-Available Reports. Accessed April 10, 2021 from: <https://www.maine.gov/dhhs/mecdc/public-health-systems/data-research/vital-records/mcr/reports/index.htm>
- Siegel DA, Li J, Henley SJ, et al. Geographic Variation in pediatric cancer incidence—United States, 2003–2014. *MMWR Morb Mortal Wkly Rep*. 2018;67(25):707–13.
- Centers for Disease Control and Prevention. BRFSS Prevalence & Trends Data. Maine. Topic: Current Smoker

13. Li J, Thompson TD, Miller JW, Pollack LA, Stewart SL. Cancer incidence among children and adolescents in the United States, 2001–2003. *Pediatrics*. 2008;121(6):e1470–77.
14. Henley SJ, Ward EM, Scott S, et al. Annual report to the nation on the status of cancer, part I: National Cancer Statistics. *Cancer*. 2020;126(10):2225–49.
15. Roadmaps Chra. Explore Health Rankings. 2022. Accessed January 28, 2022 from: <https://www.countyhealthrankings.org/explore-health-rankings>
16. Centers for Disease Control and Prevention. Leading cancer cases and deaths, all races and ethnicities, Male and female 2018. Cancer statistics at a glance. 2022. Accessed February 23, 2022 from: <https://gis.cdc.gov/Cancer/USCS/#/AIAGlance/>
17. Hosgood III HD, Farah C, Black CC, Schwenn M, Hock JM. Spatial and temporal distributions of lung cancer histopathology in the state of Maine. *Lung Cancer*. 2013;82(1):55–62.
18. Liu B, Quan X, Xu C, et al. Lung cancer in young adults aged 35 years or younger: a full-scale analysis and review. *J Cancer*. 2019;10(15):3553–9.
19. Raglan O, Kalliala I, Markozannes G, et al. Risk factors for endometrial cancer: an umbrella review of the literature. *Int J Cancer*. 2019;145(7):1719–30.
20. McVeigh UM, Tepper JW, McVeigh TP. A review of breast cancer risk factors in adolescents and young adults. *Cancers (Basel)*. 2021;13(21):5552.
21. Bleyer A, Barr R, Hayes-Lattin B, et al. The distinctive biology of cancer in adolescents and young adults. *Nat Rev Cancer*. 2008;8(4):288–98.
22. Tricoli JV, Bleyer A. Adolescent and young adult cancer biology. *Cancer J*. 2018;24(6):267–74.
23. U.S. Prevention Services Task Force. A & B Recommendations. 2022. Accessed November 29, 2021 from: <https://www.uspreventiveservicestaskforce.org/uspstf/recommendation-topics/uspstf-and-b-recommendations>
24. Barr RD, Ries LAG, Trama A, et al. A system for classifying cancers diagnosed in adolescents and young adults. *Cancer*. 2020;126(21):4634–59.
1. United States Department of Health and Human Services registries/list.html
2. National Cancer Institute. Surveillance, Epidemiology, and End Results Program. List of SEER Registries. 2021. Accessed February 26, 2022 from: <https://seer.cancer.gov/registries/list.html>
3. Fritz A, Constance P, Jack A, et al. International Classification of Diseases for Oncology, 3rd ed. 2000. World Health Organization. Accessed February 26, 2022 from: <https://apps.who.int/iris/handle/10665/42344>
4. Centers for Disease Control and Prevention and National Cancer Institute. United States Cancer Statistics-Incidence: 1999–2017, WONDER Online Database. Accessed December 15, 2020 from: <https://wonder.cdc.gov/>
5. Maine.gov. Arsenic. Accessed February 22, 2022 from: <https://www.maine.gov/dhhs/mecdc/public-health-systems/health-and-environmental-testing/arsenic.htm>
6. Maine.gov. Radon & Home Sales. Division of Environmental and Community Health. Accessed February 22, 2022 from: <https://www.maine.gov/dhhs/mecdc/environmental-health/radon/radon-homes-sales.htm#:~:text=Radon%20can%20be%20found%20in,that%20indoor%20radon%20comes%20>
7. Centers for Disease Control and Prevention. CDC WONDER. Accessed August 27, 2020 from: <https://wonder.cdc.gov/>
8. Bleyer A, O'Leary M, Barr R, Ries LAG. Cancer epidemiology in older adolescents and young adults 15 to 29 years of age, including SEER Incidence and Survival: 1975–2000. Vol. NIH Pub. No. 06-5767. Bethesda, MD: National Cancer Institute; 2006.
9. Scott AR, Stolzhus KC, Tchetebebi LT, et al. Trends in Cancer Incidence in US Adolescents and Young Adults, 1973–2015. *JAMA Netw Open*. 2020;3(12):e2027738.
10. National Cancer Institute. Surveillance, Epidemiology, and End Results Program. List of SEER Registries. 2021. Accessed February 26, 2022 from: <https://seer.cancer.gov/registries/list.html>
11. United States Department of Health and Human Services registries/list.html
12. Fritz A, Constance P, Jack A, et al. International Classification of Diseases for Oncology, 3rd ed. 2000. World Health Organization. Accessed February 26, 2022 from: <https://apps.who.int/iris/handle/10665/42344>

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